Bogdan "Bo" Alexandru Stoica

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Research Interests

My research focuses on the reliability, efficiency, and security of software systems, with an emphasis on improving their operation, maintenance, and quality assurance lifecycles at scale. To this end, I build tools to help software engineers better write, deploy, and reason about code, enabling them to identify faults more efficiently. I am interested in program analysis, efficient code instrumentation, performance reasoning, and artificial intelligence approaches that support software engineering tasks.

Education

University of Chicago — Chicago, IL, United States

Ph.D. in Computer Science (Area: Systems and Software Engineering) — September 2019 – August 2025

Dissertation: Principles and Tool Support for Detecting Latent Bugs in Modern Software Systems (link)
Advisor: Prof. Shan Lu
Mentors: Prof. Haryadi Gunawi, Prof. Kexin Pei, Dr. Suman Nath and Dr. Madan Musuvathi

M.Sc. in Computer Science (Area: Systems and Software Engineering) — September 2019 – June 2022

Dissertation: Exposing Memory Ordering Bugs Efficiently with Active Delay Injection
Advisor: Prof. Shan Lu
Mentors: Prof. Haryadi Gunawi, Dr. Suman Nath and Dr. Madan Musuvathi

École Polytechnique Fédérale de Lausanne (EPFL) — Lausanne, Switzerland

Graduate Coursework (Area: Systems and Software Engineering) — September 2014 – August 2015

Mentors: Prof. Vikram S. Adve, Prof. Bernard M. E. Moret and Prof. Viktor Kunčak

École Polytechnique Fédérale de Lausanne (EPFL) — Lausanne, Switzerland

M.Sc. in Communication Systems (Area: Information Security) — September 2011 – January 2014

Thesis: Robust Web Content Evaluation
Advisor: Prof. Karl Aberer

University of Bucharest (UB) — Bucharest, Romania

B.Sc. in Computer Science (Area: Algorithms and Software Security) — October 2008 – June 2011

Thesis: On Intrusion Detection Systems
Advisor: Prof. Adrian Atanasiu

Awards and Honors

Distinguished Artifact Evaluator Award (EuroSys'25)	2025
Eckhardt Graduate Fellowship (University of Chicago)	2019 – 2024
Teaching Assistant Award for Outstanding Service (EPFL)	2016 – 2017
Excellence Fellowship for Master's Studies (EPFL)	2011 – 2013
Excellence Scholarship for Bachelor's Studies (UB)	2008 – 2011
Bronze Medal, International Pluridisciplinar Olympiad in Informatics (Tuymaada), Russian Federation	2007

Peer-Reviewed Conference Publications

[C4] **Who Watches the Watchers? On the Reliability of Softwarizing Cloud Application Management.** Jiawei Tyler Gu, Zhen Tang, Yiming Su, Bogdan Alexandru Stoica, Xudong Sun, William X. Zheng, Yue Zhang, Akond Rahman, Chen Wang, and Tianyin Xu. *In Proceedings of the 23rd USENIX Symposium on Operating Systems Design and Implementation (NSDI), Renton, WA, May 2026. (To appear).* NSDI '26

[C3] **Synthesizing Performance Constraints for Evaluating and Improving Code Efficiency.** Jun Yang, Cheng-Chi Wang, Bogdan Alexandru Stoica, Kexin Pei. *In Proceedings of the 39th Conference on Neural Information Processing System (NeurIPS), San Diego, CA, Dec 2025. (To appear).* NeurIPS '25

[C2] **If At First You Don't Succeed, Try, Try, Again...? Insights and LLM-informed Tooling for Detecting Retry Bugs in Software Systems.** Bogdan Alexandru Stoica*, Utsav Sethi*, Yiming Su, Cyrus Zhou, Shan Lu, Jonathan Mace, Madan Musuvathi, and Suman Nath (*equal contribution). *In Proceedings of the 30th ACM Symposium on Operating Systems Principles. Austin, TX, US. November 2024.* SOSP '24

[C1] **Waffle: Exposing Memory Ordering Bugs Efficiently with Active Delay Injection.** Bogdan Alexandru Stoica, Shan Lu, Madan Musuvathi, and Suman Nath. *In Proceedings of the 18th ACM SIGOPS European Conference on Computer Systems. Rome, Italy. May 2023.* EuroSys '23

Peer-Reviewed Workshop, Short, and Poster Publications

[W1] **Wok: Statistical Program Slicing in Production.** Bogdan Alexandru Stoica, Swarup K. Sahoo, James R. Larus, Vikram S. Adve. *In Proceedings of the 41st ACM/IEEE International Conference on Software Engineering. Montreal, QC, Canada. May 2019.* ICSE '19

Pre-prints and technical reports

[P1] **The Discreet Charm of the Bugeoisie.** Ji young Kim, Jana Dragovic, Alessandro Botta, Rithwanul Islam, Alaa Mohamad, Karim Sharaf, Sejuti Sharmin Siddiqui, Divyanshi Joshi, Harini Anand, Nurjemal Saryyeva, Shubham Chapagain, Saad Nasir, Darko Marinov, and Bogdan Alexandru Stoica. *Under submission, January 2026.* .

[P2] **An Empirical Study of Failures in Softwarized Cloud Application Management.** Jiawei Tyler Gu, Zhen Tang, Yiming Su, Bogdan Alexandru Stoica, Xudong Sun, William X. Zheng, Yue Zhang, Akond Rahman, Chen Wang, and Tianyin Xu. *Under submission. January 2026* .

[P3] **SemRep: Code Transformation with Semantics-Preserving Representations.** Weichen Li, Jiamin Song, Bogdan Alexandru Stoica, Arav Dhoot, Gabriel Ryan, Shengyu Fu, and Kexin Pei. *Under submission, January 2026* .

[P4] **Statistical Program Slicing: a Hybrid Slicing Technique for Analyzing Deployed Software.** Bogdan Alexandru Stoica, Swarup K. Sahoo, James R. Larus, and Vikram S. Adve. *arXiv:2201.00060, December 2021.* `https://arxiv.org/abs/2201.00060` arXiv '21

Employment

University of Illinois at Urbana-Champaign — Urbana, IL, United States

Postdoctoral Research Associate — August 2025 – present

Hosts: Prof. Tianyin Xu and Prof. Darko Marinov

I work at the intersection of Systems, Software Engineering, Programming Languages, and Artificial Intelligence (AI), designing techniques that help software engineers better write, deploy, and reason about their code. I build tools that combine traditional software testing and monitoring techniques with modern AI approaches.

University of Chicago — Chicago, IL, United States

Research Assistant September 2019 – August 2025
Advisor: Prof. Shan Lu

I developed automated fault detection techniques for large-scale systems implemented in a wide range of programming languages (C, C++, C#, Java, and Python). I designed bug-finding tools that combine program analysis, fault injection, code instrumentation, and AI-guided code comprehension to isolate and expose correctness and performance defects.

Google Sunnyvale, CA, United States
PhD Software Engineering Intern (Core ML Engineering) June 2023 – September 2023
Mentors: Gloria Shen and Dr. Ilya Kavalerov

I prototyped a pipeline that helps software engineers pinpoint regression bugs in Google's ML infrastructure more efficiently. The key functionality of the pipeline is matching character-level differences between consecutive builds to high-order program structures (e.g. class, package, etc.) which are later traced back to the relevant tests exercising these code changes. The prototype is currently being integrated with TensorFlow's build framework.

Meta Seattle, WA, United States
Research Intern (Systems & Infrastructure, Profiling) June 2022 – October 2022
Mentors: Nathan Slingerland and Jacie Fan

I developed a rules-based engine that identifies inefficient C++ code patterns at scale by analyzing billions of low-level execution profiles collected across Cloud services. The tool is currently being integrated with Meta's existing code efficiency analysis protocols to help developers pinpoint difficult-to-find performance bottlenecks.

Microsoft Research Redmond, WA, United States
Research Collaborator January 2020 – June 2020
Mentors: Dr. Suman Nath and Dr. Madan Musuvathi

I explored a series of bug diagnosis techniques that integrate with existing software testing frameworks. I investigated fault injection strategies to help expose difficult-to-reproduce bugs in distributed applications. This exploratory work led to Waffle, a push-button fault injection tool that helps developers trigger memory ordering concurrency bugs (see [C1]) and to Wasabi, a LLM-informed fault injection tool that helps developers expose bugs in retry mechanisms (see [C2]).

Microsoft Research Redmond, WA, United States
Research Intern July 2018 – September 2018
Mentors: Dr. Weidong Cui and Dr. Ben Niu

I designed and implemented a C++ tool for tracing heap memory management requests to help program state recovery during offline execution replay. This extended Windows Debugger's reverse debugging engine by increasing the current recovery rate and allowing it to replay longer execution traces.

École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland
Research Assistant September 2015 – December 2018
Mentors: Prof. Vikram S. Adve, Prof. Bernard M. E. Moret, and Dr. Swarup K. Sahoo

I designed techniques to help developers analyze their code more efficiently. I prototyped a suite of C and C++ tools for scalable bug diagnosis using program analysis, efficient code instrumentation, and emerging hardware support for execution tracing.

Microsoft Prague, Czech Republic
Software Development Engineer (Automated Testing Infrastructure) February 2014 – August 2014
Mentor: Travis Merkel

I helped develop automated testing frameworks (C, C++ and Python) for the Skype tool chain. I used these to design performance testbeds which identified several critical memory leaks and buffer overflow bugs.

Bitdefender Labs Bucharest, Romania
Software Development Engineering Intern (R&D) June 2012 – September 2012
Mentor: Teodor Stoenescu

I developed a stand-alone, secondary SSL certificate validation tool for the Bitdefender Anti-virus suite (C/C++). Parts of my code were integrated in the 2013 release of the software.

Bitdefender Labs Bucharest, Romania
Software Development Engineer (R&D) May 2010 – August 2011
Mentors: Mihai Chiriac and Teodor Stoenescu

I developed several modules of a new Anti-virus suite for virtual environments (C/C++). I focused on optimizing network traffic processing and implemented a multi-layer cache which increased scanning throughput by 50%.

Talks

[T9] **The Missing Machinary in Artifact Evaluation.**
The 6th Chameleon User Meeting (NSF Symposium). Boulder, CO, US. Invited Talk. April 2026

[T8] **If At First You Don't Succeed, Try, Try, Again...? Insights and LLM-informed Tooling for Detecting Retry Bugs in Software Systems.**
SOSP'24. Austin, TX, US. Conference Talk. November 2024

[T7] **Weaving Large Language Models into the Bug Finding Pipeline: Challenges and Opportunities.**
PACMI'24. Austin, TX, US. Invited Talk. November 2024
The 5th Chameleon User Meeting. Atlanta, GA, US. Conference Talk. November 2024

[T6] **Understanding, Characterizing and Exposing Deeply-Nested Bugs in Large-Scale Systems.**
University of Illinois at Urbana-Champaign. Urbana, IL, US. Invited Talk. October 2024

[T5] **Artifact Reproducibility as a Classroom Tool.**
ACM REP'24. Rennes, France. Invited Tutorial Talk. June 2024

[T4] **Waffle: Exposing Memory Ordering Bugs Efficiently with Active Delay Injection.**
EuroSys'23. Rome, Italy. Conference Talk. May 2023

[T3] **Failure Diagnosis with Hardware Support**
Imperial Collage London, London, UK. Seminar. January 2019
University of Illinois at Urbana-Champaign. Urbana, IL, US. Invited Talk. September 2018

[T2] **Exploring Hardware Data Logging on Modern CPUs**
Microsoft Research. Redmond, WA, US. Seminar. September 2018

[T1] **Modern Hardware and OS Support for Efficient Execution Tracing**
University of Zurich. Zurich, Switzerland. Invited Talk. December 2017

Professional Service

Conference Chairing and Steering Committe Service
Artifact Evaluation Chair for the ACM Conference on AI and Agentic Systems (CAIS) 2026
Artifact Evaluation Co-Chair for the European Conference on Computer Systems (EuroSys) 2026
Artifact Evaluation Vice-Chair for the Intl. Conf. on Parallel Arch. and Compilation Techniques (PACT) 2026

Program Committee Service
European Conference on Computer Systems (EuroSys) 2026
ACM Conference on AI and Agentic Systems (CAIS) 2026
The Annual Conference on Machine Learning and Systems (MLSys) ERC 2026
Practical Adoption Challenges of ML for Systems (PACMI@SOSP) 2025

Artifact Evaluation Committee Service
European Conference on Computer Systems (EuroSys) 2025

Symposium on Operating Systems Principles (SOSP)	2023
The Annual Conference on Machine Learning and Systems (MLSys)	2023
Symposium on Operating Systems Design and Implementation (OSDI)	2022
European Conference on Computer Systems (EuroSys)	2022
Intl. Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)	2022
Symposium on Operating Systems Principles (SOSP)	2021
Intl. Conf. on Programing Languages Design and Implementation (PLDI)	2019
Symposium on Principles and Practice of Parallel Programming (PPoPP)	2018, 2019

Teaching Experience

Co-instructor

UChicago, CMSC-33200: Topics in Operating Systems (graduate)	2024

Guest Lecturer

UChicago, CMSC-33200: Topics in Operating Systems (graduate)	2023

Teaching Assistant

UChicago, CMSC-14300: Systems Programming (undergraduate), lead teaching assistant	2023
UChicago, CMSC-22001: Software Construction (undergraduate), lead teaching assistant	2022
UChicago, MPCS-52030: Operating Systems (graduate), teaching assistant	2020
UChicago, CAPP-30122: Computer Science with Applications II (graduate), teaching assistant	2020
UChicago, MPCS-55001: Algorithms (graduate), teaching assistant	2019
EPFL, CS-173: Digital Systems Design (EPFL, undergraduate), lead teaching assistant	2018
EPFL, CS-207: Systems Oriented Programming (EPFL, undergraduate), lead teaching assistant	2015, 2017
EPFL, CS-250: Algorithms (EPFL, undergraduate), teaching assistant	2015, 2016, 2017
EPFL, CS-450: Advanced Algorithms (EPFL, graduate), teaching assistant	2013, 2014, 2016
EPFL, CS-150: Discrete Structures (EPFL, undergraduate), teaching assistant	2013

Research Mentoring

M9. **Ji young Kim**. UIUC (PhD)	2025 – now
M8. **Jun Yang**. UChicago (PhD)	2024 – 2025
M7. **Casper Wang**. National Taiwan University (BSc)	2024 – 2025
M6. **Zahra Nabila Maharani**. Dian Nuswantoro University (BSc)	2023 – 2025
M5. **Wordyka Nainggolan**. Del Insitute of Technology (BSc)	2024 – 2025
M4. **Shuang Liang**. Ohio State (BSc)	2023 – 2024
M3. **Yiming Su**. UChicago (BSc) → UIUC (PhD). Publications: [C2]	2023 – 2024
M2. **Cyrus Zhou**. UChicago (BSc). Publications: [C2]	2023 – 2024
M1. **Rizky Ramadhana P. K**. Bandung Institute of Tech (BSc) → Virginia Tech (MSc)	2022 – 2023